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MILANO 1863

DIPARTIMENTO
DI MECCANICA

Surface modelling of a bicycle helmet for virtual impact testing (EN 1078)

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Introduction

01



Objectives:

- Starting from the scanned (photogrammetry) helmet, represented by a mesh, model both manually and automatically the object using NURBS surfaces.
- Use the final models (Manual and Auto) to find, through FEM, their shock absorbing capacity following the EN 1078 standard, for which the helmet is certified.
- Compare the results.

Headform modelling

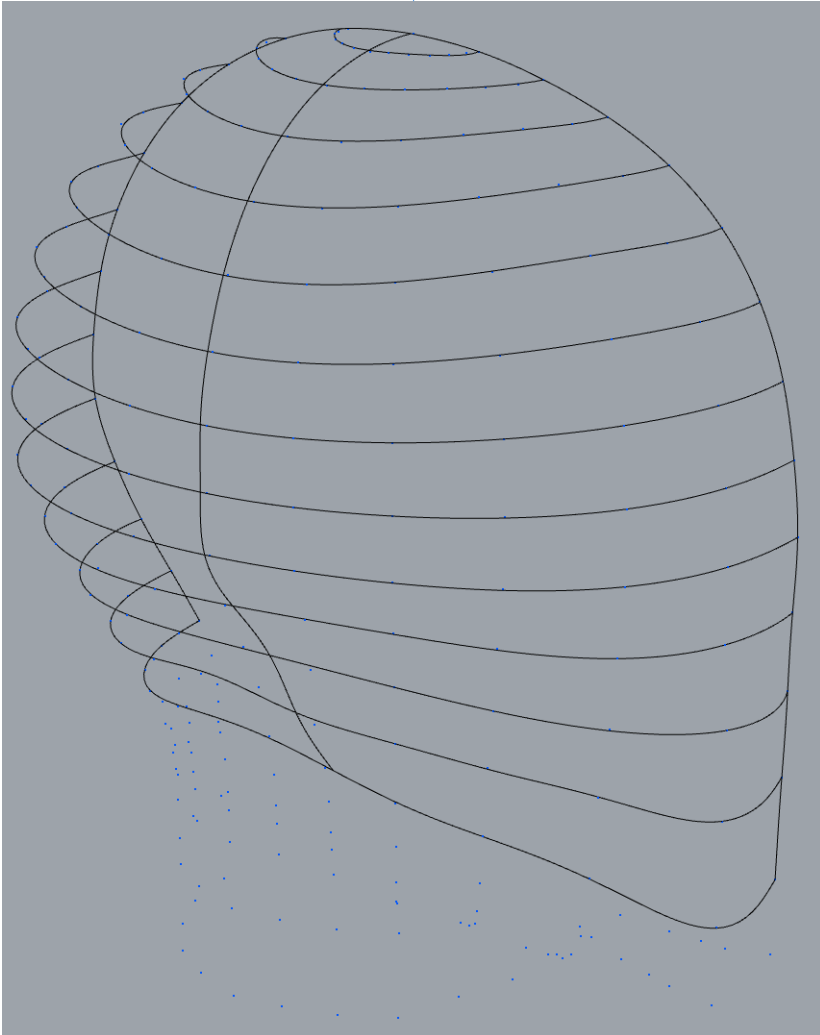
02

1. Measured internal circumference of the helmet: 600 mm.
2. From EN 1078 (Table 1) standard this value corresponded to size designation 605.
3. Spherical coordinates of half headform (size 605) are provided in EN 960 standard.
4. Spherical to rectangular coordinates:

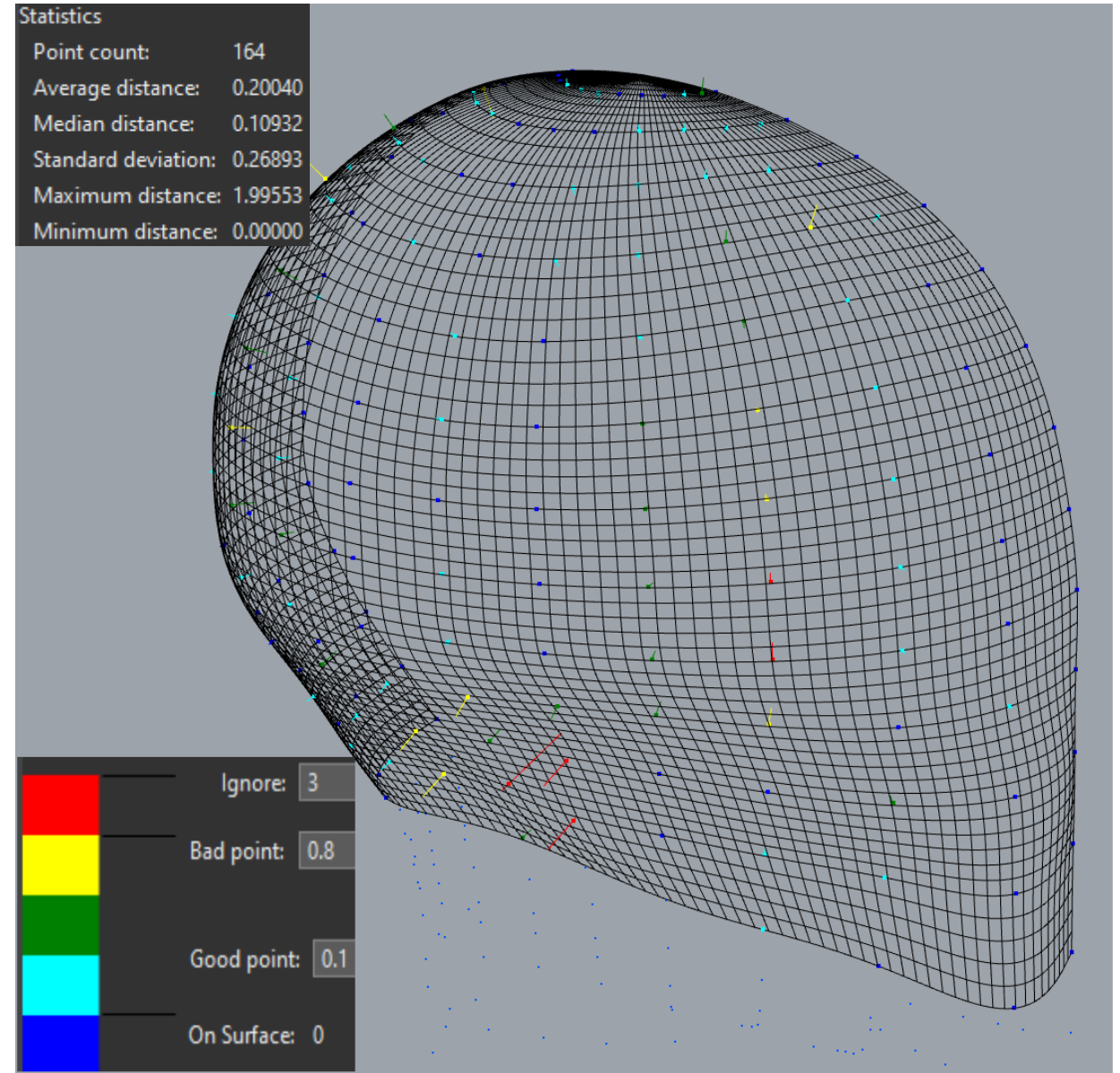
```
void computeCoord3D(int angV, int angH, double radius, Coord3& coords){  
    const double angVrad {deg2rad(angV)};  
    const double angHrad {deg2rad(angH)};  
    coords.x = radius*std::cos(angVrad)*std::sin(angHrad);  
    coords.y = radius*std::cos(angVrad)*std::cos(angHrad);  
    coords.z = radius*std::sin(angVrad);  
}
```

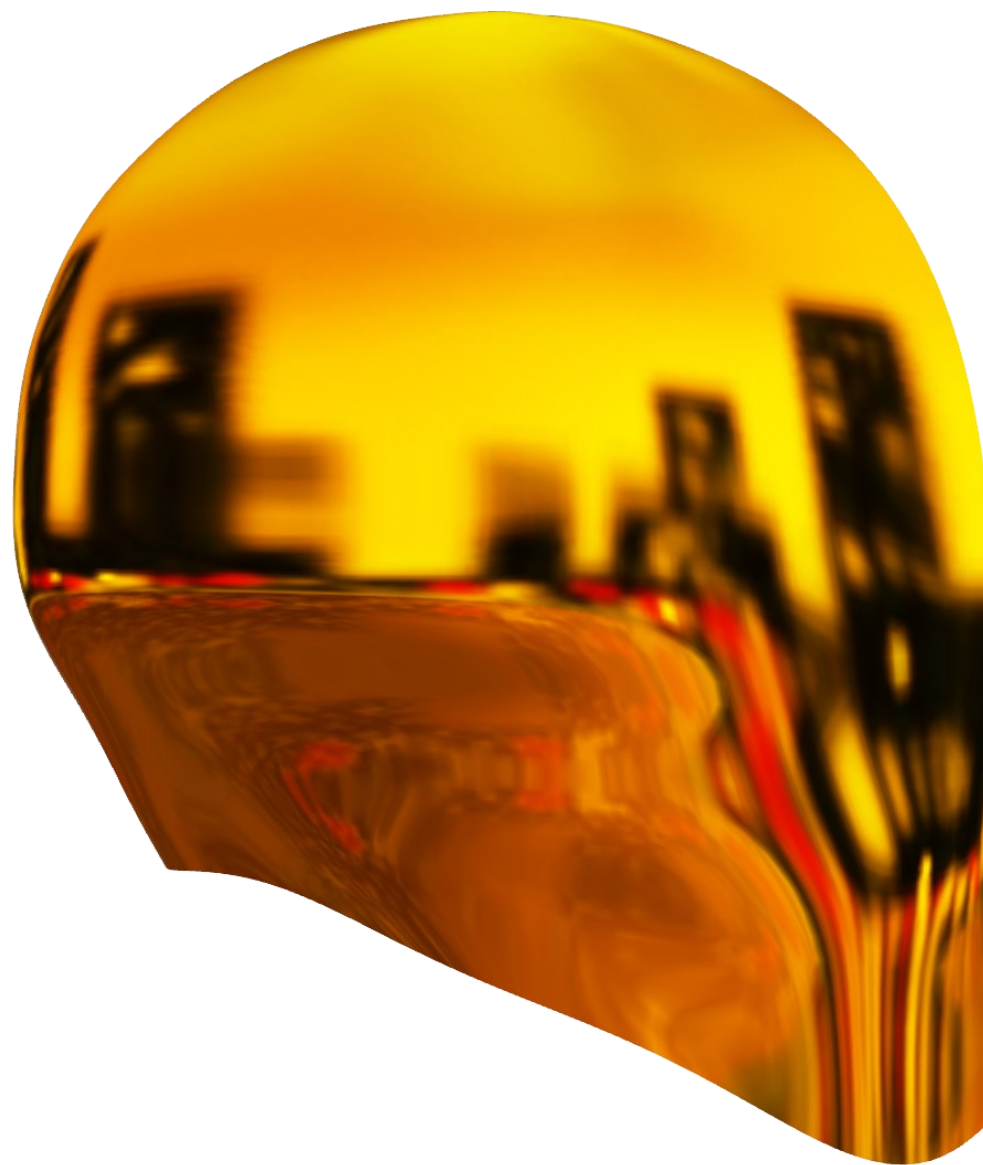
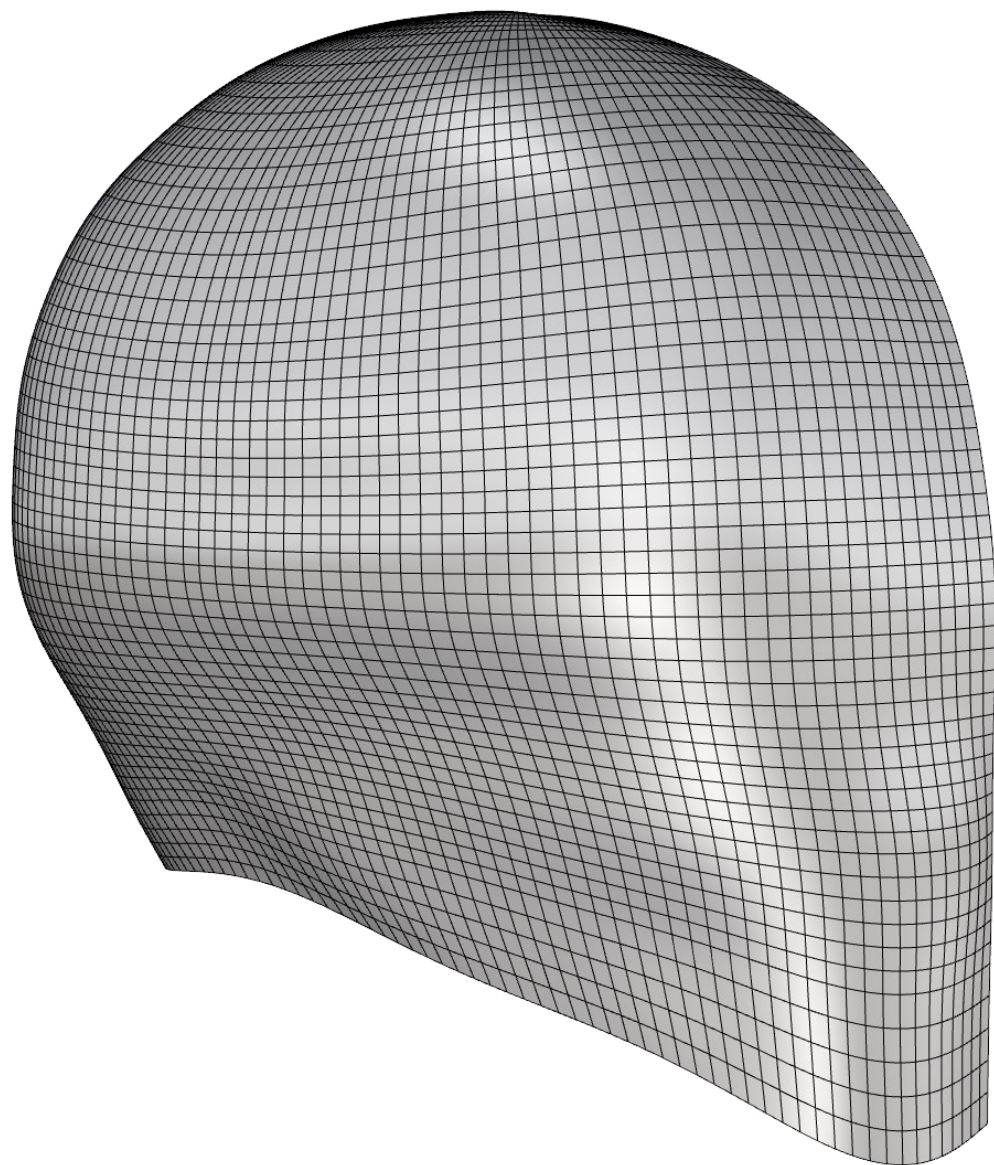
5. The rectangular 3D coordinates written to a .csv file are imported in Rhino as a point cloud.
6. For each set of points at the same angle from the horizontal reference plane (down to -40°):
 - InterpCrv of degree 5 and chord-length (negligible difference between alternatives) parametrization for knot spacing.
 - Rebuild the curve for improved fairness changing only the point/knot count of the curve with maximum deviation $\leq 0.8 \text{ mm}$.
7. From the point at the top of the headform an InterpCrv (deg. 5 and chord-length parametrization) was used to interpolate the end points and middle points of the previous constructed curves.

Obtaining the following
structured network of curves



NetworkSrf





Helmet modelling

03

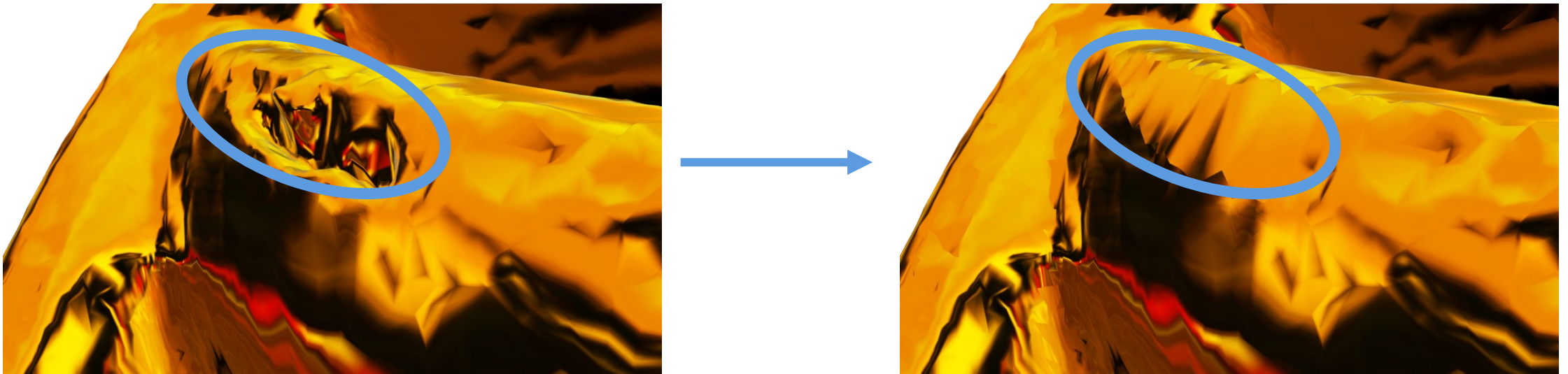
- The photos taken for photogrammetry were mainly focusing on half of the helmet, given the object's symmetry. Obtaining:



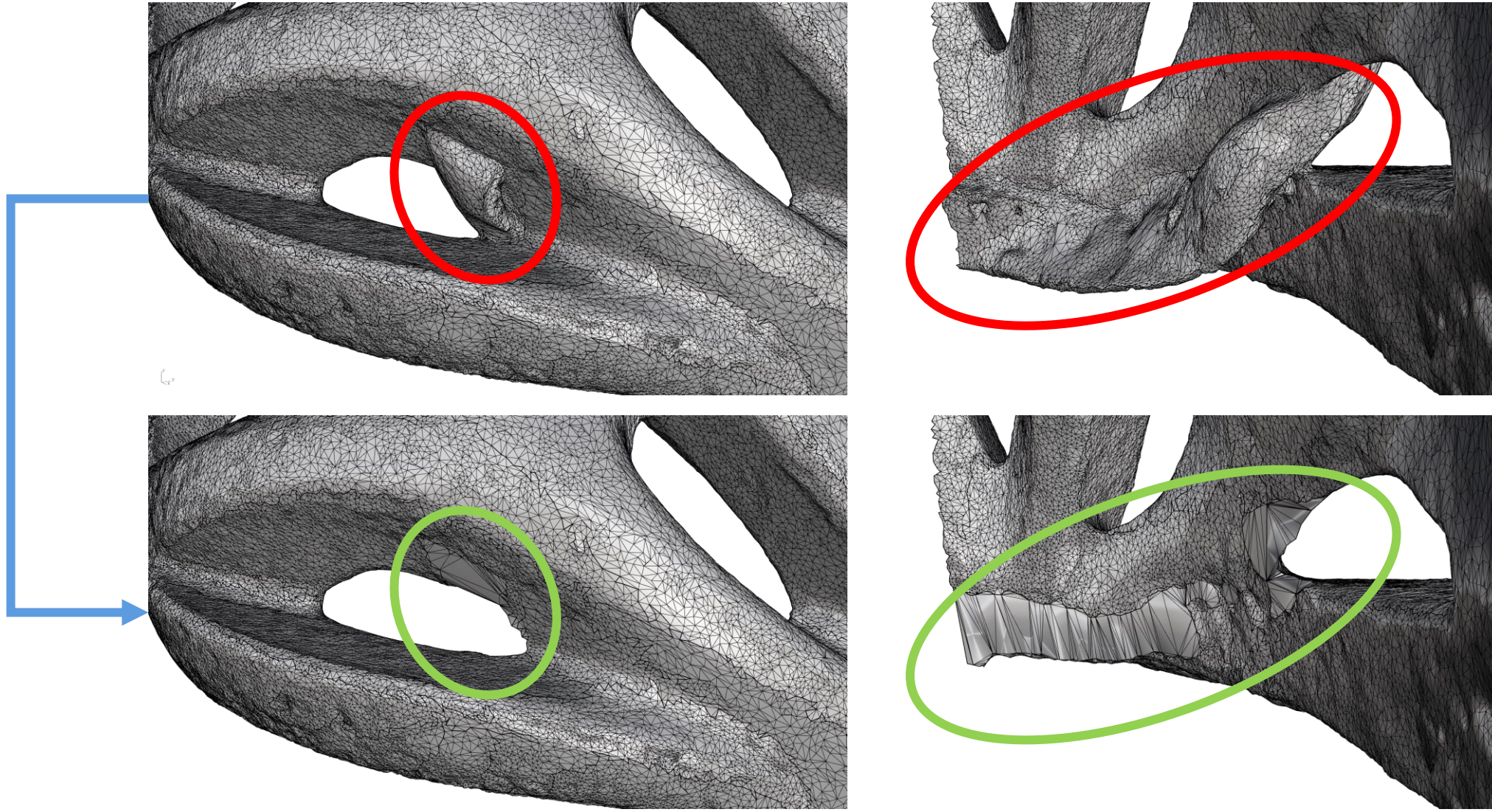
- The mesh was first oriented to match the “world” axes in Rhino.
- Scaling the mesh to match the actual helmet size:

	Inner length	Outer length
Measured experimentally	213 mm	265 mm
After mesh scaling process	213.4 mm	264.3 mm
Relative error	+0.2 %	-0.3 %

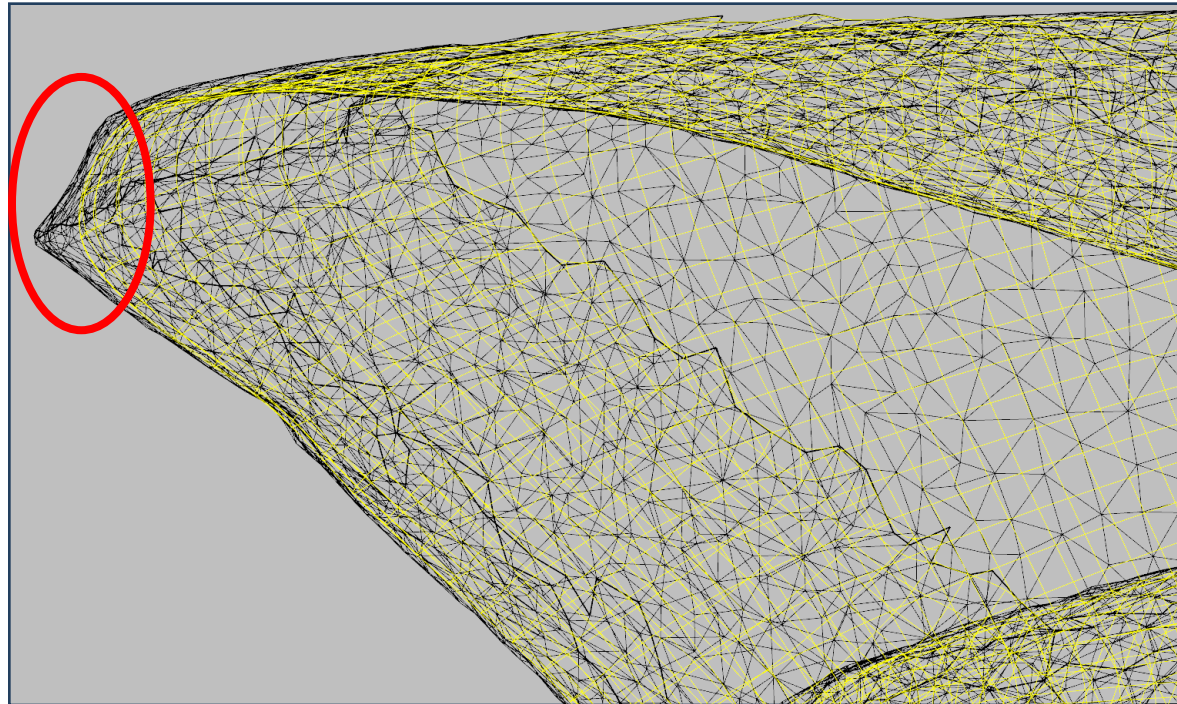
- Remove (ExtractMeshFaces) the low-quality half of the mesh from the rest of it, slightly less than half, to not lose information at the symmetry plane.
- Mesh cleaning/smoothing, starting from the largest defects:
 1. Selecting and removing the set of faces representing the defect with SelBrush
 2. Filling the hole (FillMeshHole)
 3. Changing the vertices position if the filled hole is not smooth enough.
 4. Smoothing (Smooth) the modified mesh region by averaging the vertices' position.



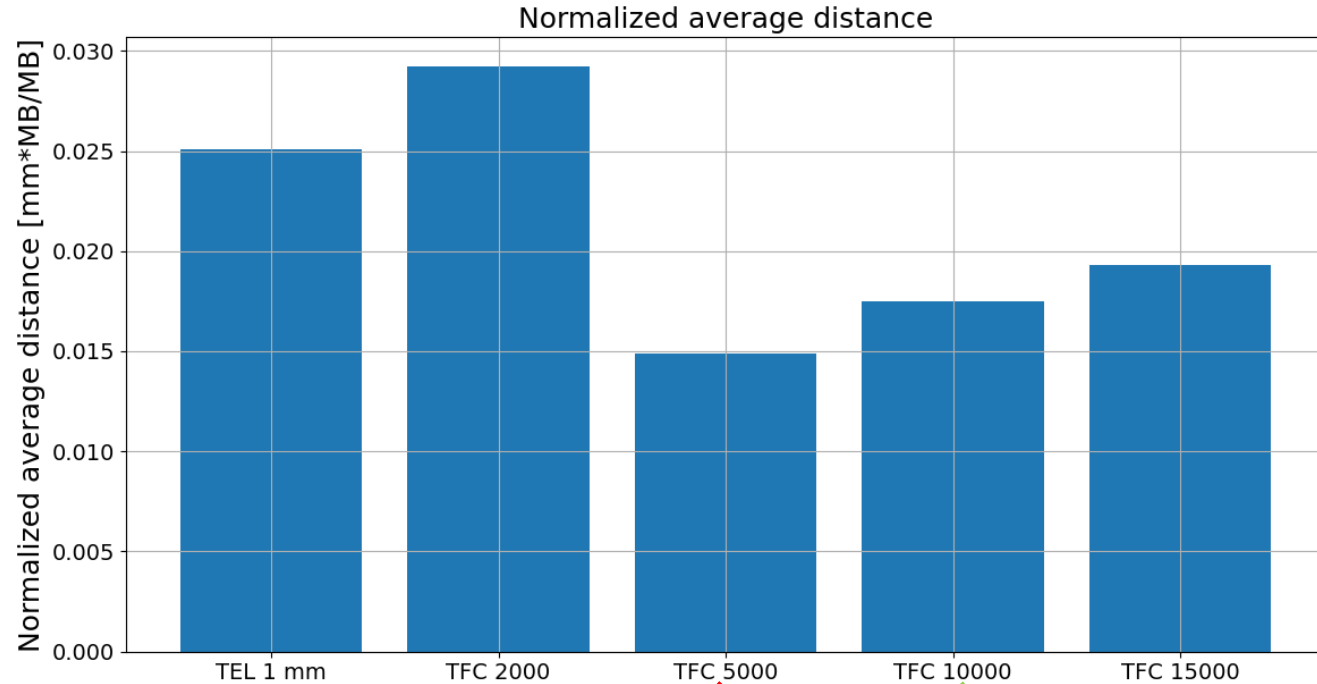
In the case of the chin-strap laces, which is considered a large defect region of the helmet (see figure below), in addition to the actions described before, also single mesh face (3DFace) generation was used together with MatchMeshEdge and Join to control the filling process.



- The mesh was then converted to quad-mesh and finally subD (QuadRemesh) with the following options:
 - Adaptive size (50%) and adaptive quad count, to improve the computational efficiency (available only with target face count).
 - Deactivating "detect hard edges", as most features are rounded.
 - Activating conversion to SubD surface.
 - No creased edges (as no hard edges are detected)
 - No sharp corners (for the same reason)
 - Interpolate SubD wasn't used to not capture sharp deviations that would cause "out-of-memory" error, even though the shrinking of the subD compared to the mesh was undesired (figure below).



➤ Different target face counts (TFC) and edge lengths (TEL) were tried and compared:



$$\bar{d}_{mesh-subD} \cdot \frac{heap_i}{heap_{max}}$$

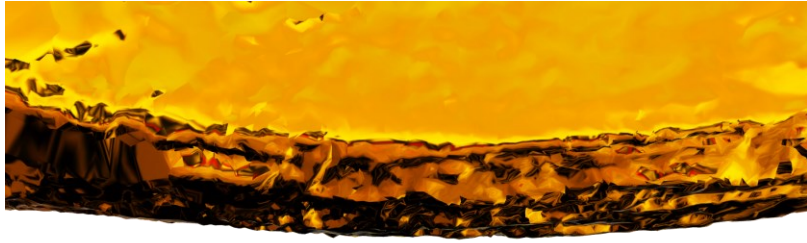
Statistics of **SubD conversion with TFC 10k**:

- Actual face count: 17034
- Average distance from mesh: 0.065 mm
- Standard deviation on dist.: 0.072 mm
- Maximum distance: 0.953 mm
- Required heap allocation: 71.5 MB

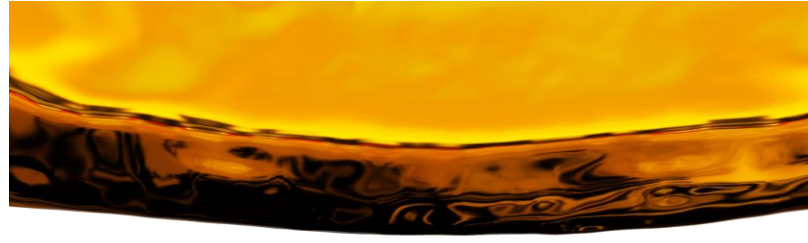
Discarded due to max.
distance $\geq 1\text{ mm}$

The **one chosen** as the
most efficient for
approximating the mesh,
which also satisfies the
max. distance constraint

SubD conversion with TFC of 10000 was chosen also because it filtered out the undesired small defects of the mesh, as shown in the figures below:



Mesh



TEL 1 mm (97311 faces)

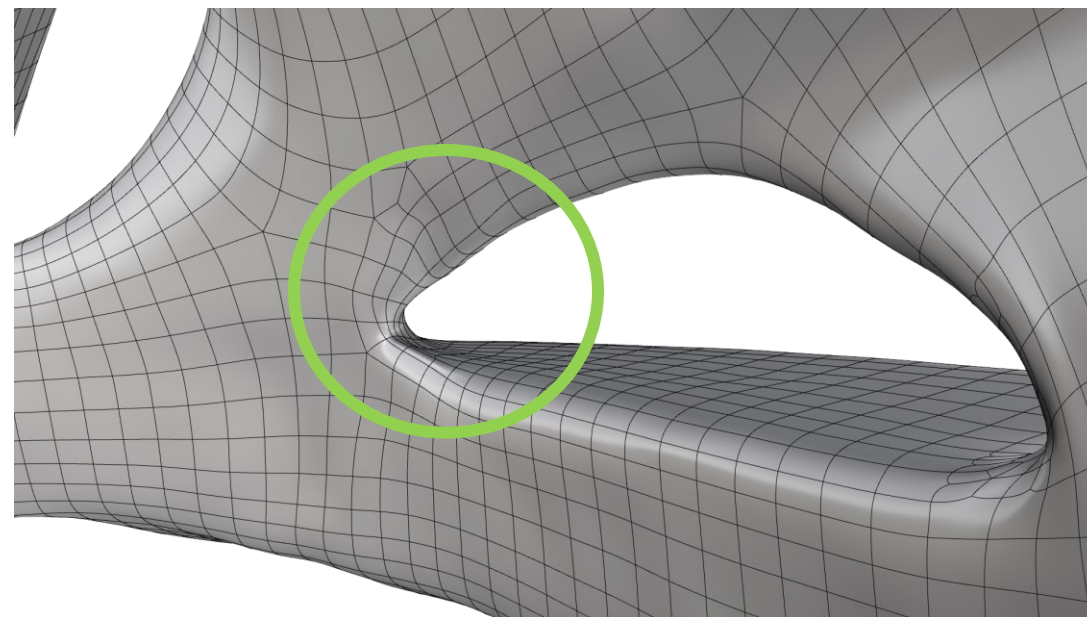
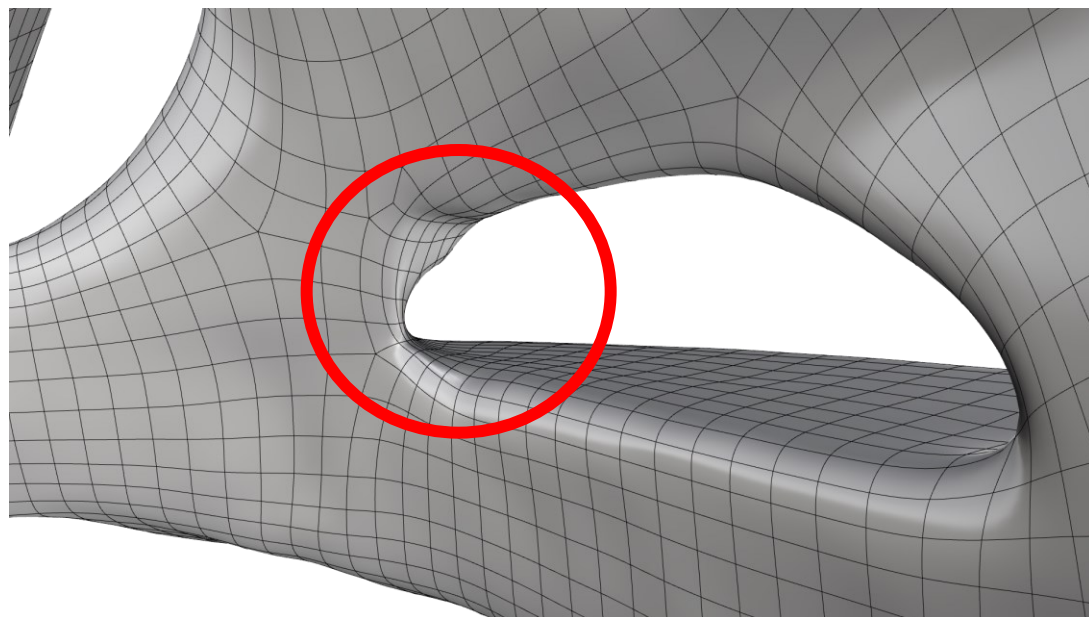
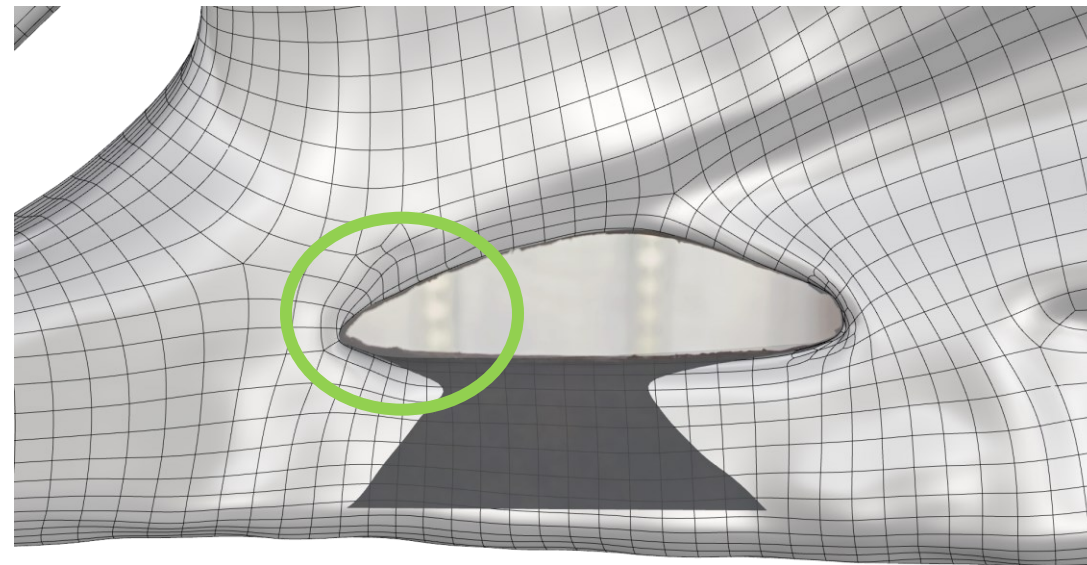
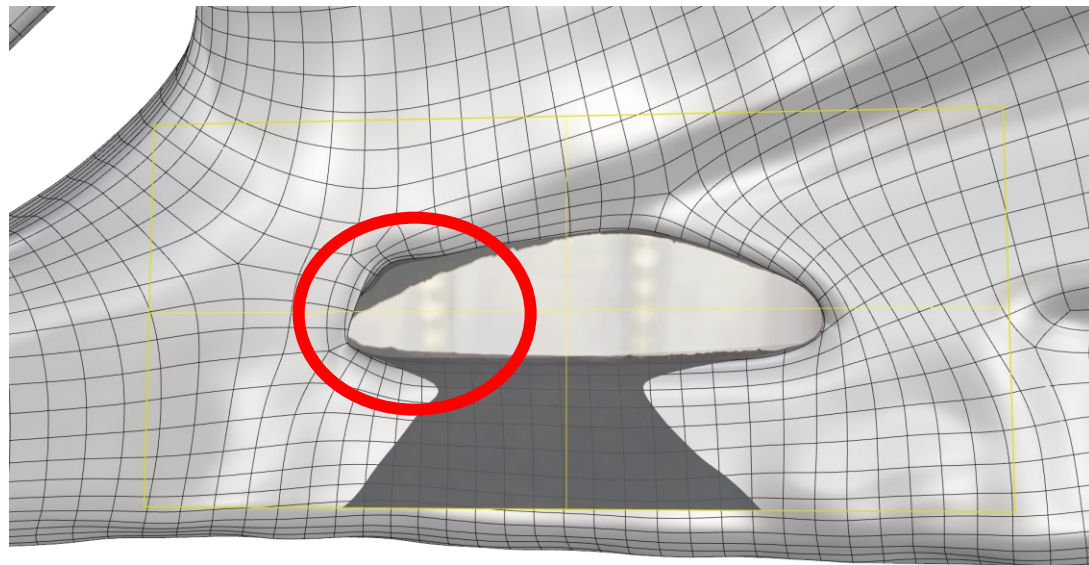


TFC 10k (17034 faces)

The chin-strap laces removal in the mesh  **temporary** representation of the helmet's hole.

Adjusted by subD surface modelling:

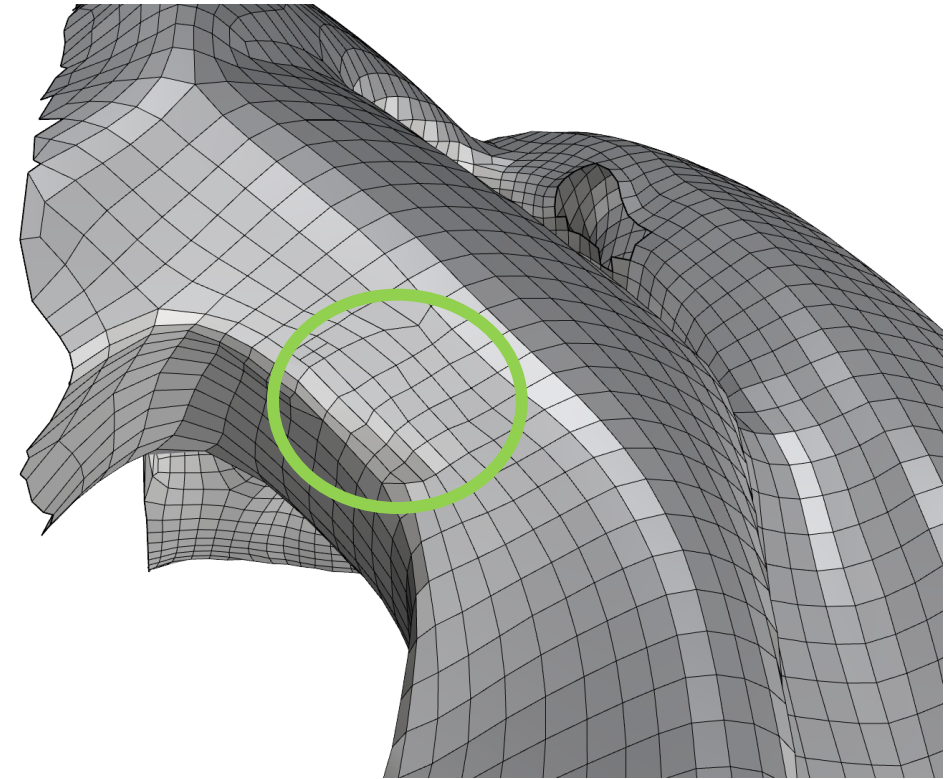
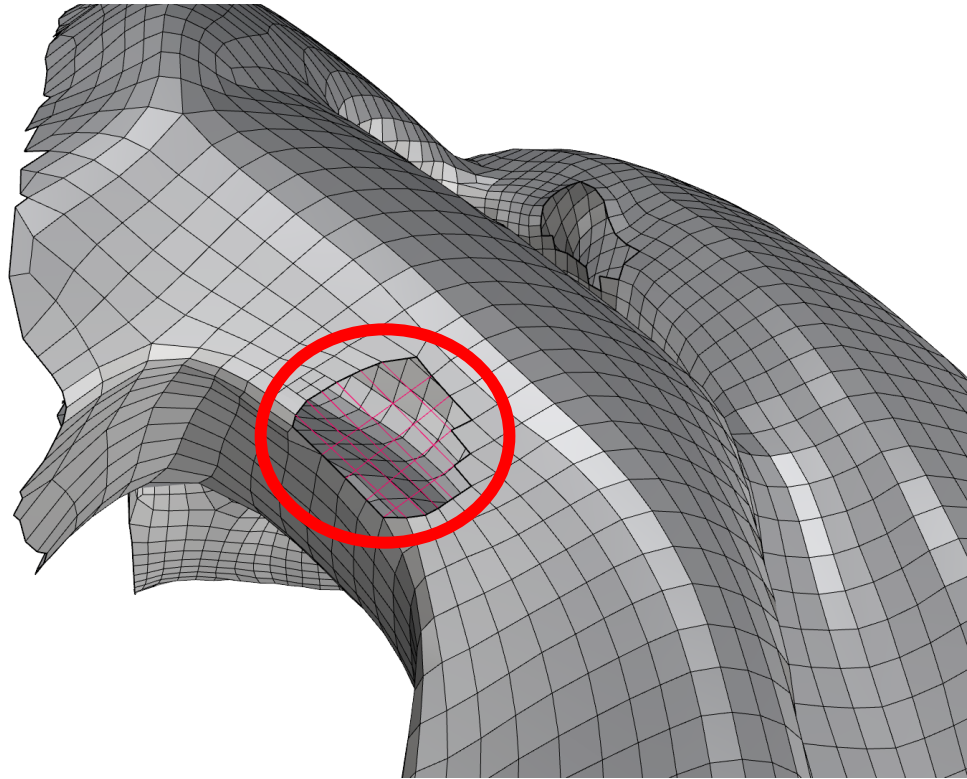
- Image of the hole + measurement used as reference.
- Sub-division where necessary.
- Moving subD vertices or edges.



Automatic extraction of NURBS surfaces

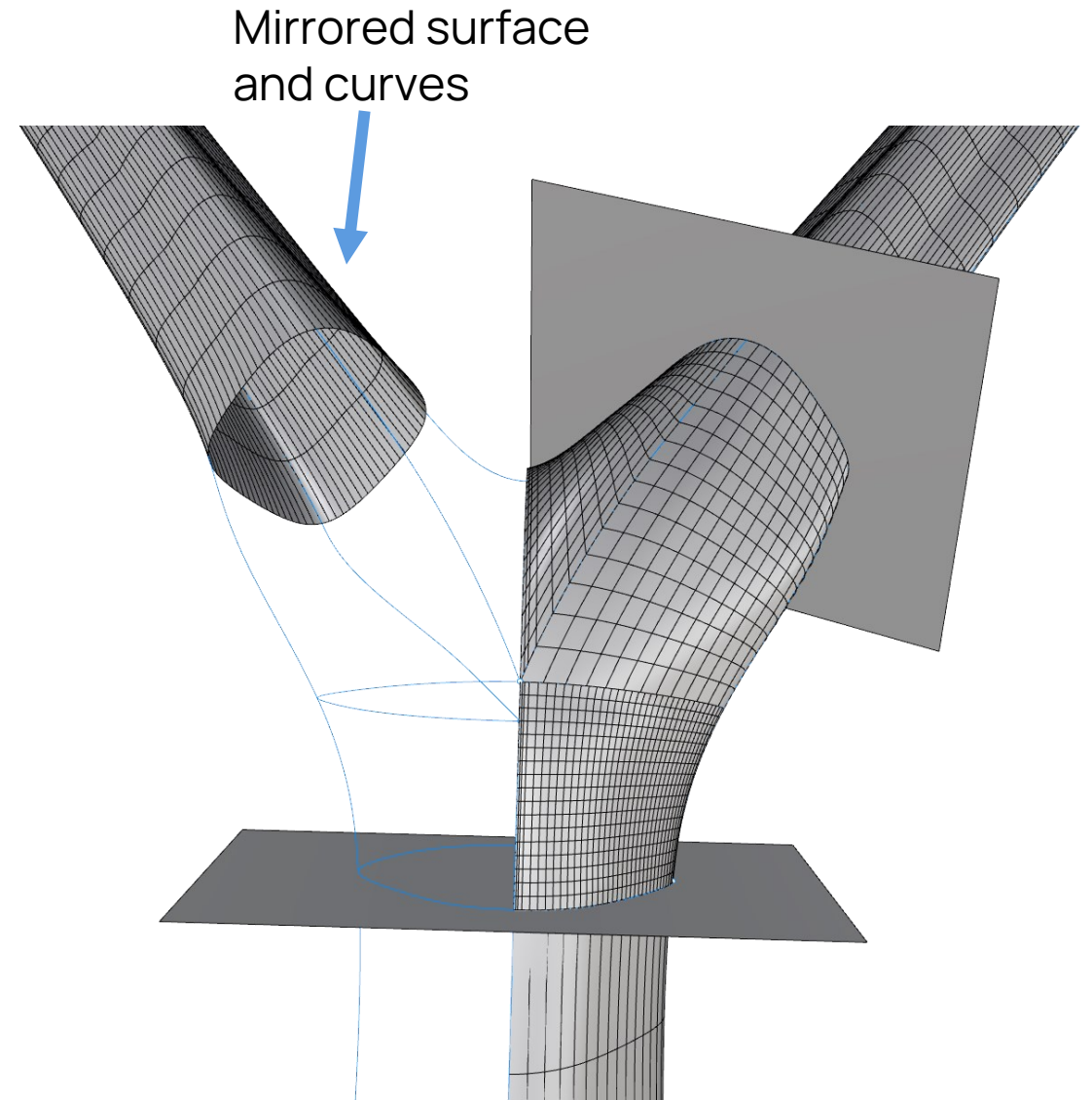
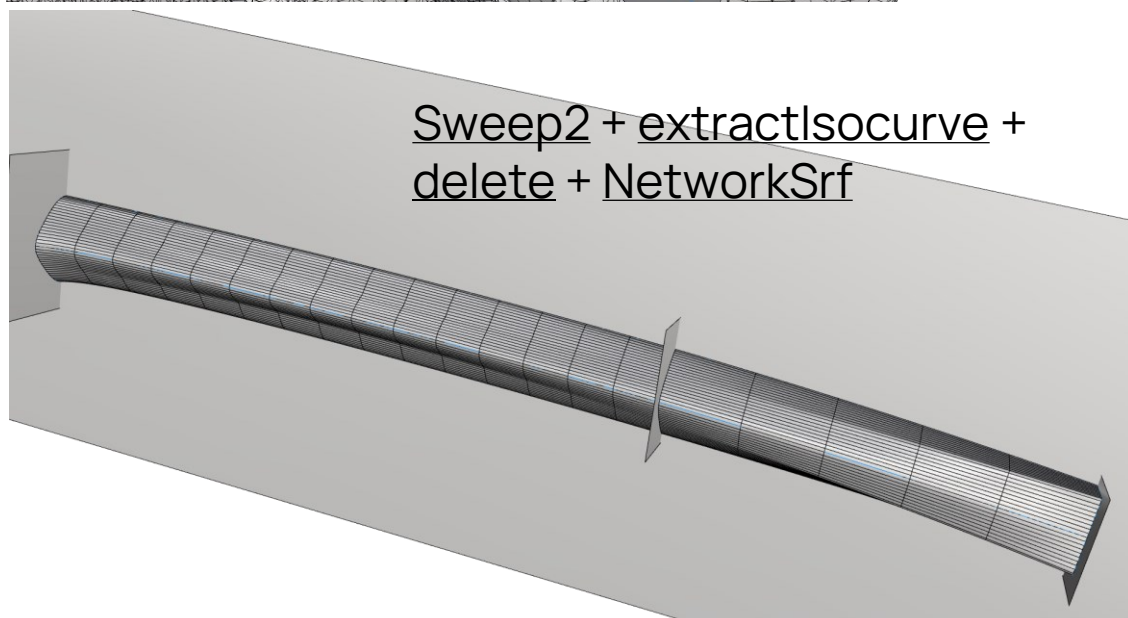
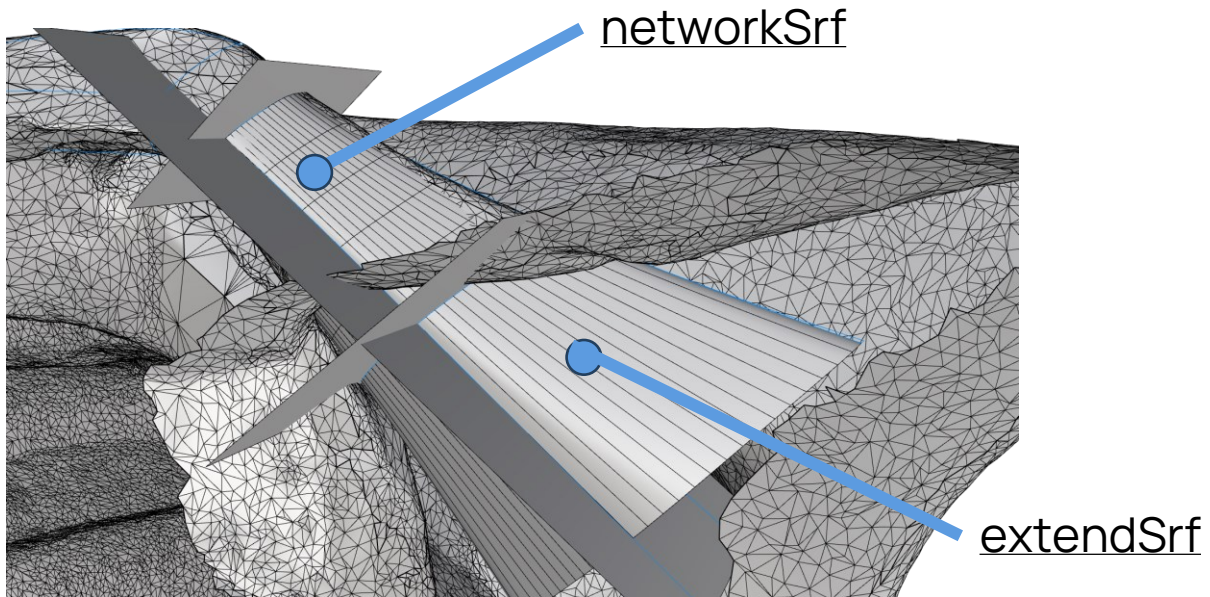
03.1

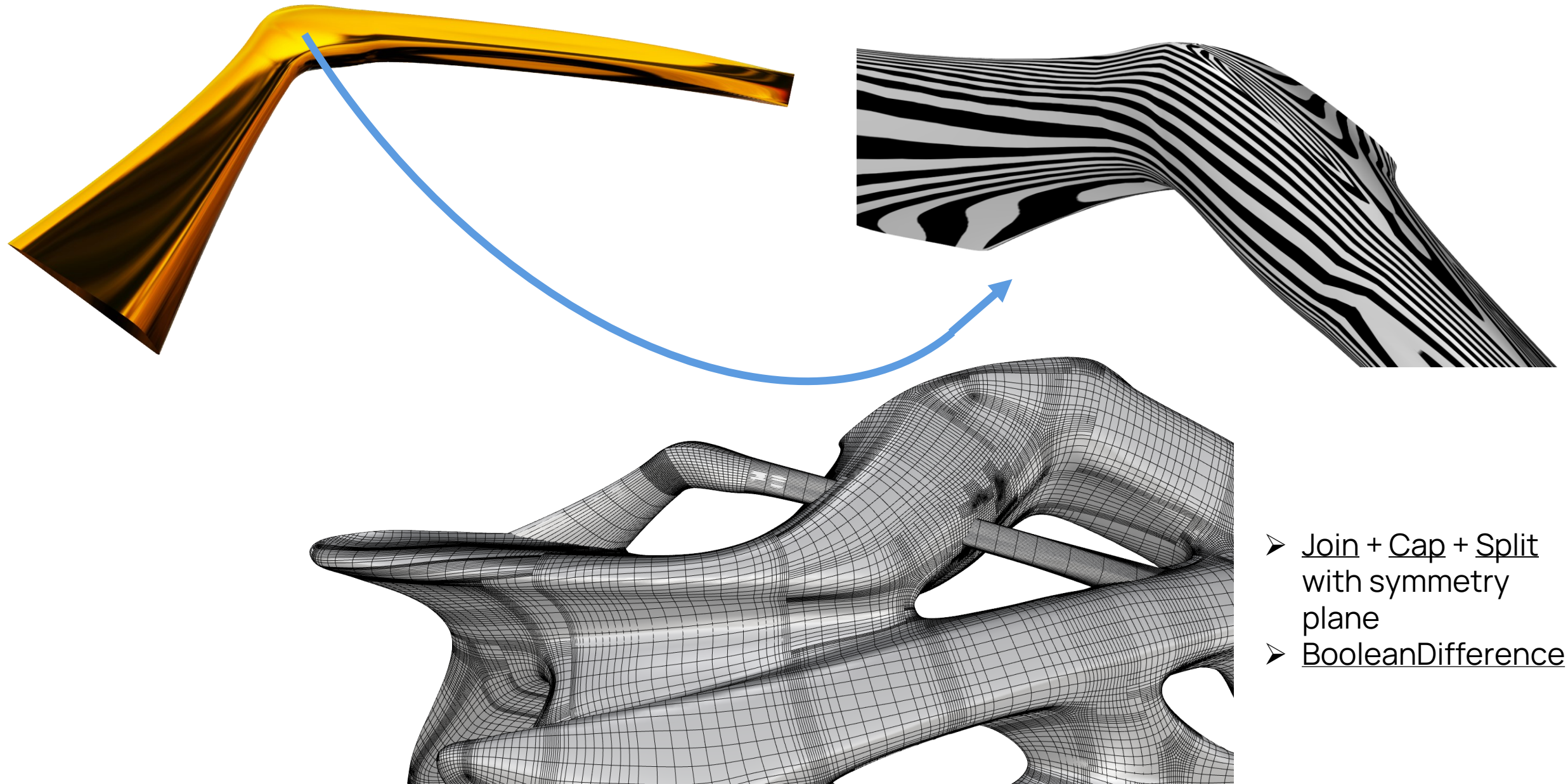
- After some other fine adjustments of the subD surface (Smooth) and the removal of the plastic insert on top (3DFace + Join or append option), the automatic conversion to NURBS (ToNURBS) surfaces was achieved successfully (default settings).



- The open NURBS polysurface was split with the symmetry plane, and vice versa. Joining them created a closed polysurface, representing half helmet (913 NURBS surfaces).

Modelling of the **plastic insert** using the mesh as reference:

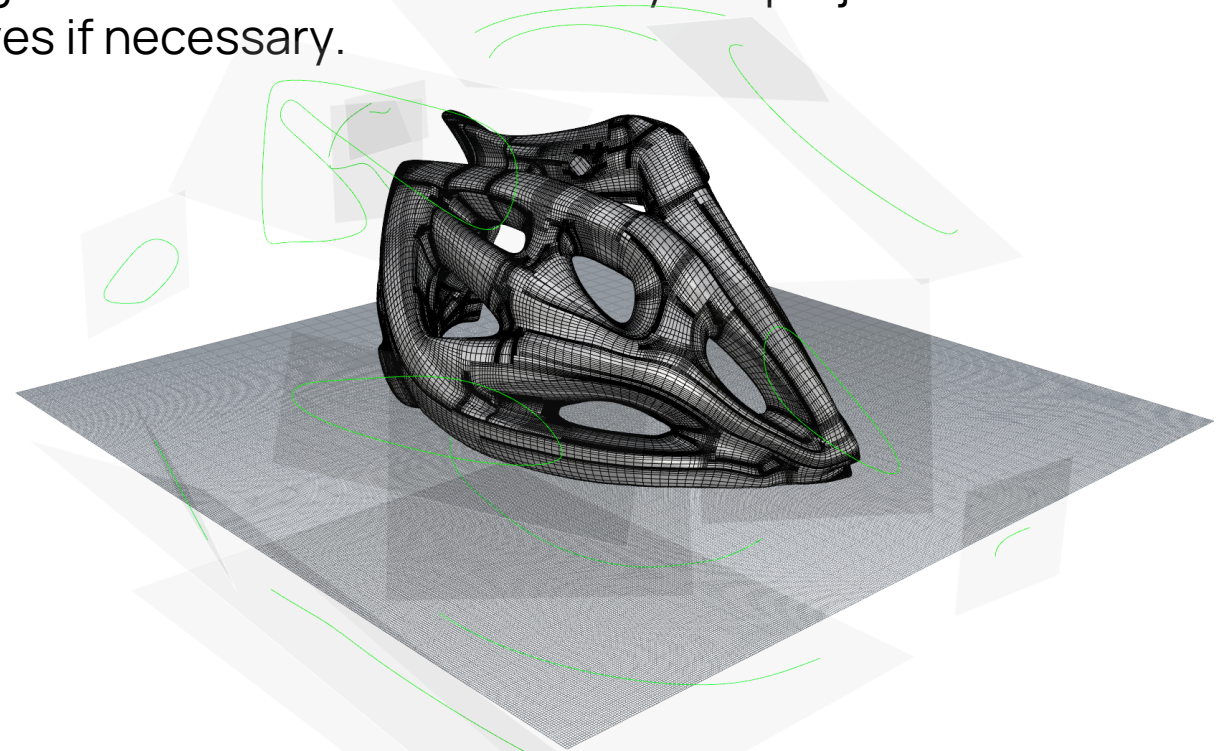
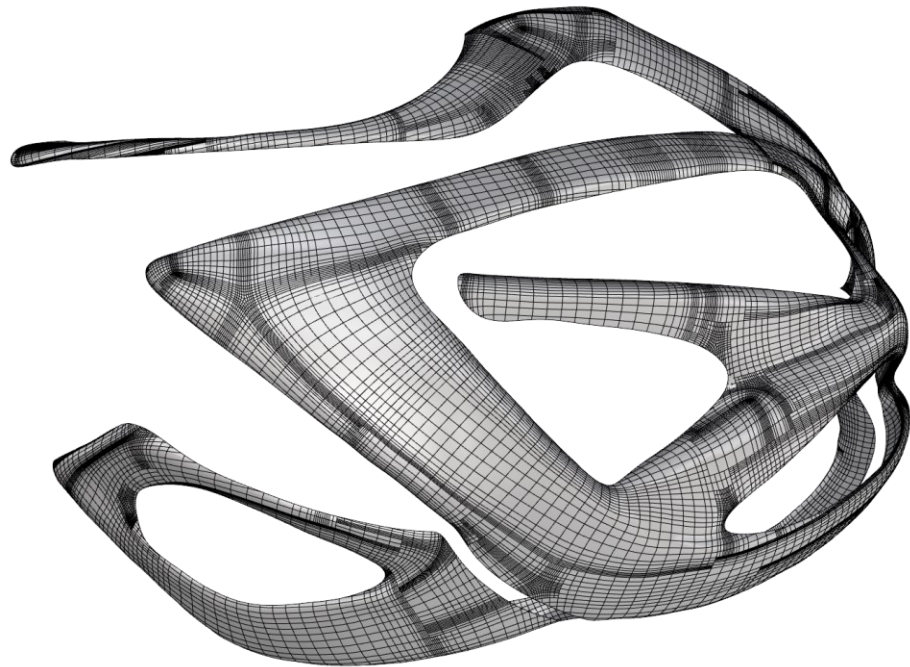




- Join + Cap + Split with symmetry plane
- BooleanDifference

For the PC outer shells, the workflow was:

- Quasi-transparent planar surface was positioned so that its planar view shows the portion of the helmet of interest.
- Cplane was set on the planar surface.
- View set to “plan”.
- Degree 5 fitting curve was drawn like it was on the surface.
- The curve was projected on the helmet surface after some modification to its control points.
- Curve was rebuilt with less control points (loosing the projection property).
- BlendCrv with other curves if necessary.
- The curve was Pull onto the polysurface, selecting all the surfaces interested by the projection.
- This curve was Joined to previously modelled curves if necessary.



Manual modelling of NURBS surfaces

03.2

Starting from the subD surface previously obtained, the modelling of the helmet surfaces was pursued considering:

- The scope/usage of the surface model: non-linear explicit dynamics with contact regions (FEM).
- The level of detail that can be represented with “coarse” finite elements.

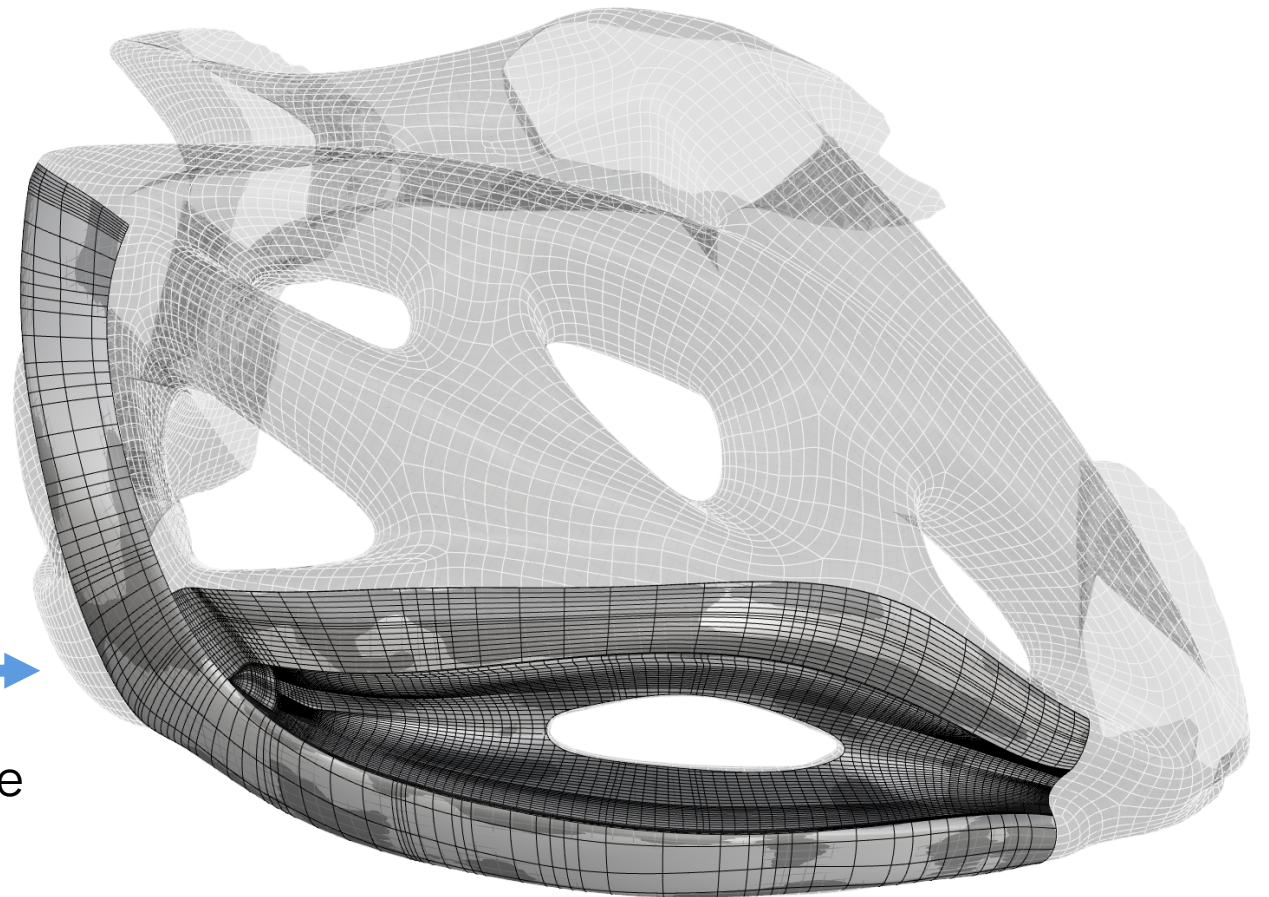
The order followed for surface modelling:

1. Outer surface + cooling holes.
2. Bottom surface.
3. Inner surface



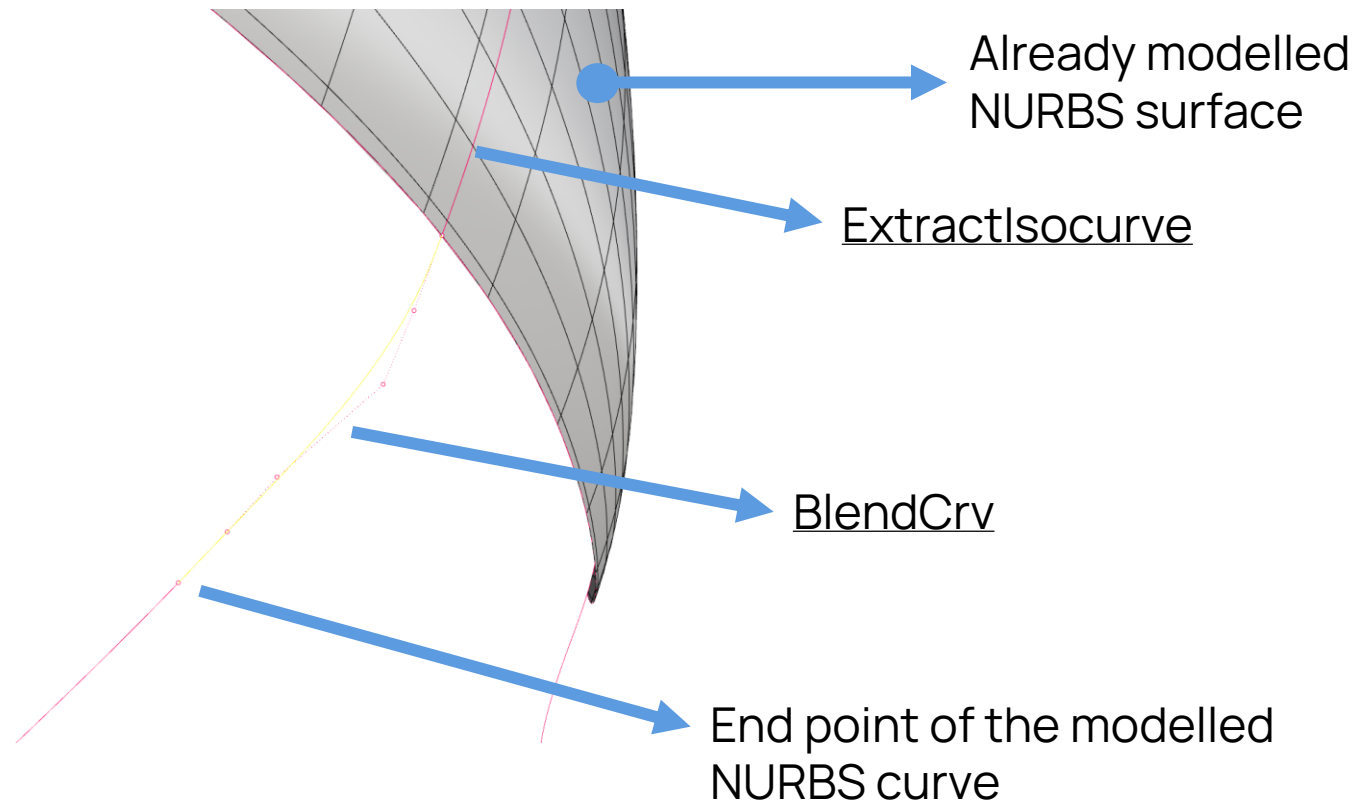
Each split into multiple regions to simplify the modelling process.

Region 0 –
outer surface



Main workflow followed:

- Planar surface used as Cplane + degree 5 fitting curve. For non-planar features, degree 5 interpolation curve + Rebuild (improving fairness).
- ExtractIsocurve + BlendCrv (G2 continuity).
- Iteratively added NURBS curves until surface quality was satisfactory.



- NetworkSrf, Sweep2 or Loft was used for the NURBS surface generation.
- G2 continuity between surfaces was imposed either from the surface generation options or using MatchSrf.

Talking about surfaces edge matching...

Abaqus FEA software requires absolute tolerance to be at least $1e-6$ units (mm) to avoid an “imprecise geometry” warning.



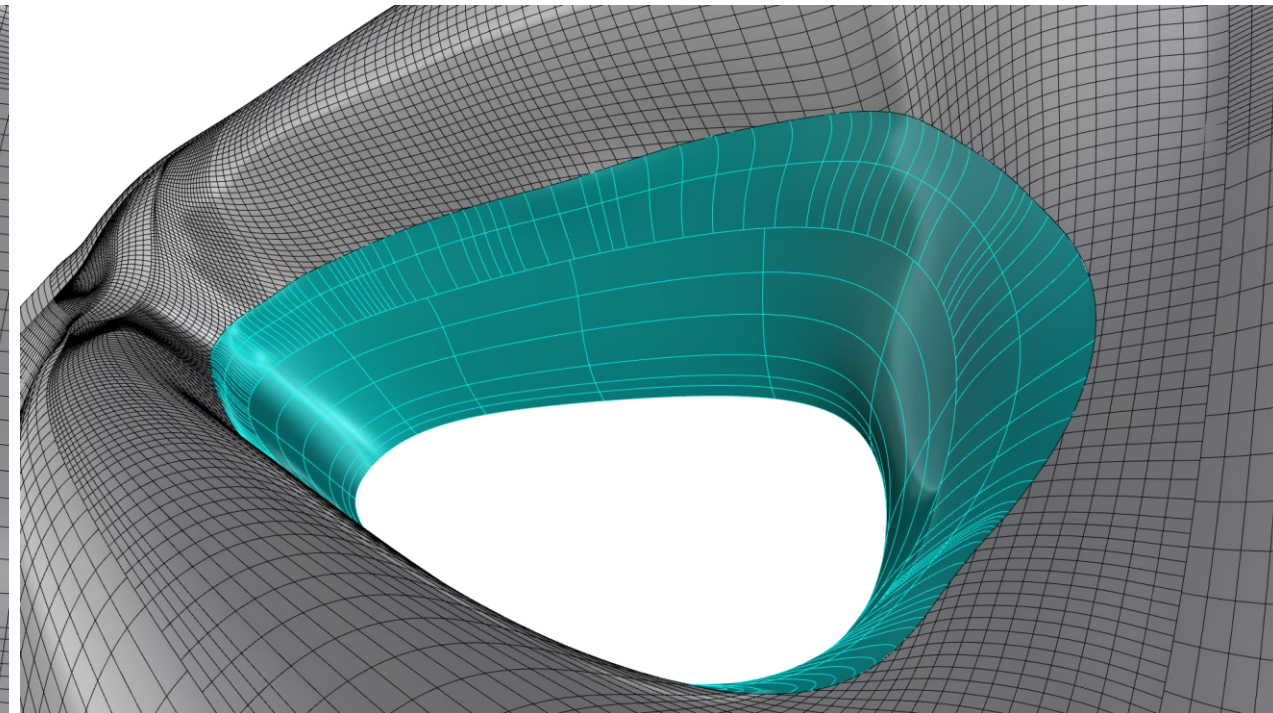
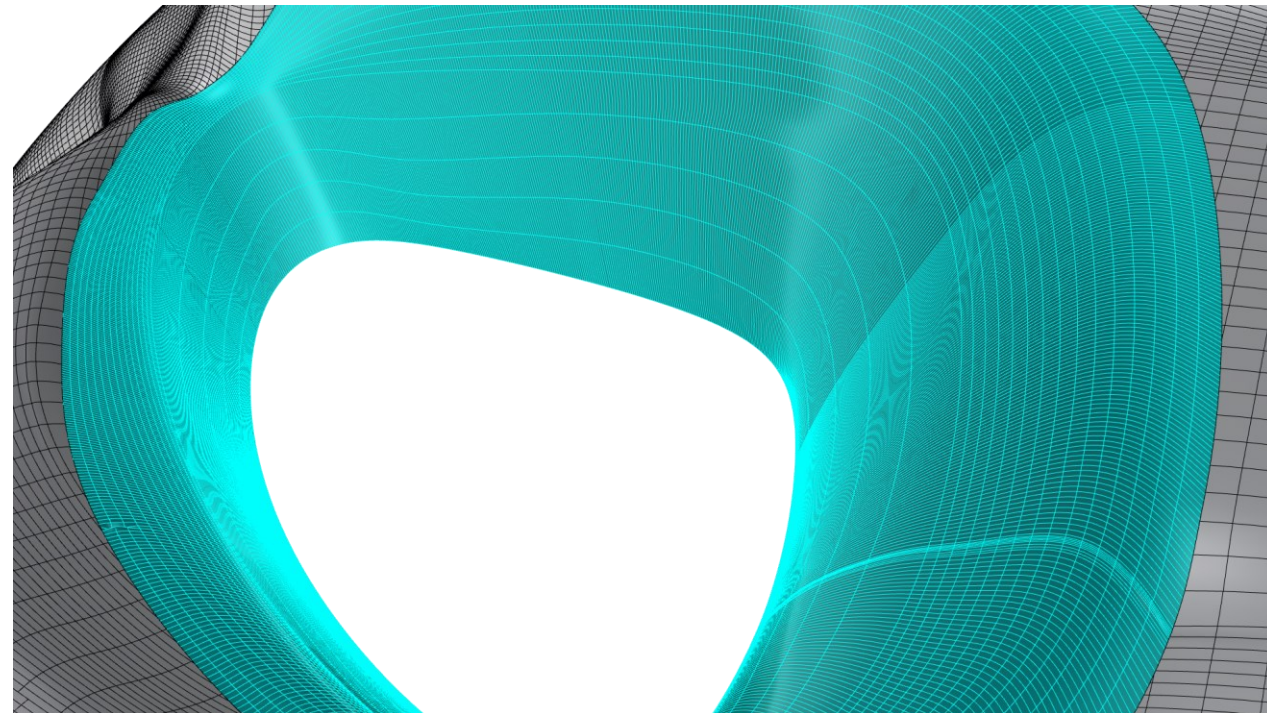
A solution could be to set Rhino's absolute tolerance to $1e-6$ units (from $1e-3$!), but this is too computationally expensive.



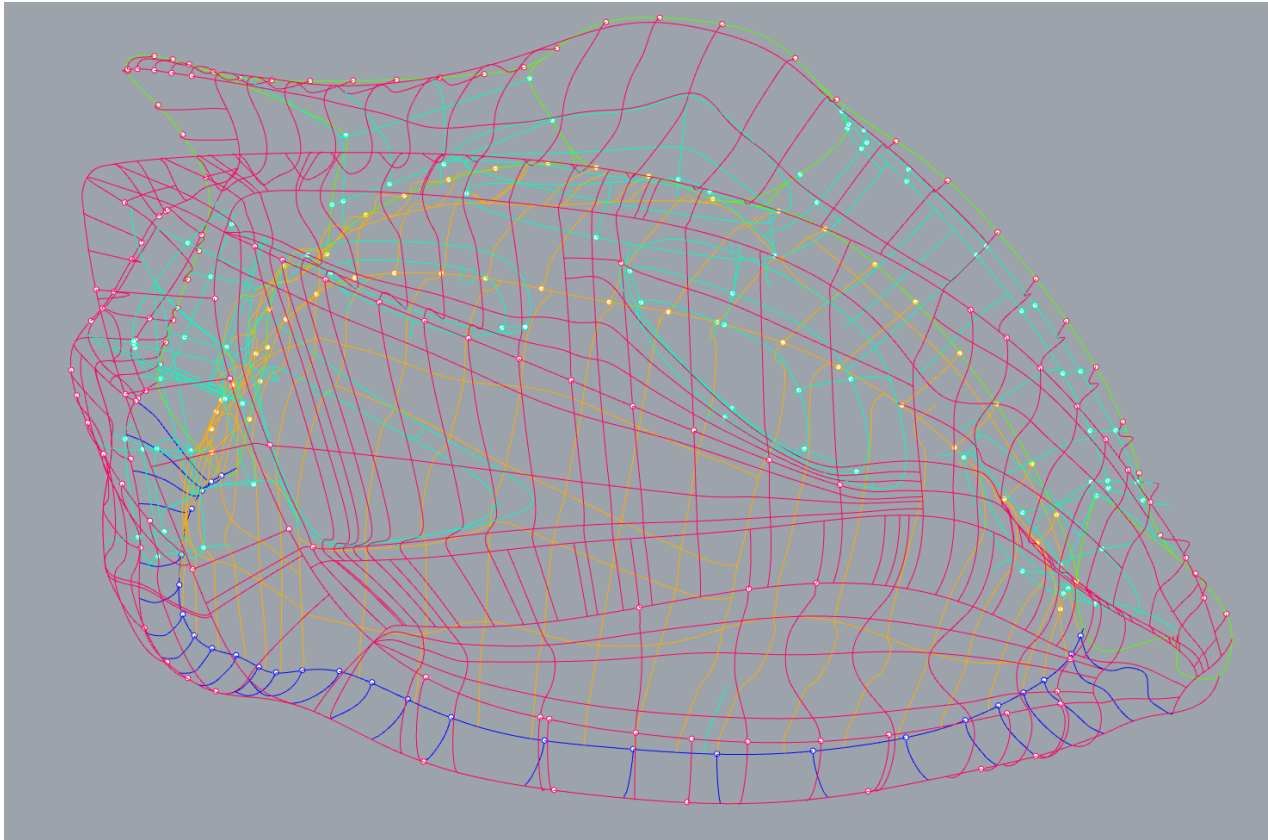
Better solution: using the default $1e-3$ mm absolute tolerance in Rhino and then improve the edge matching tolerance in Abaqus with the repair tool.

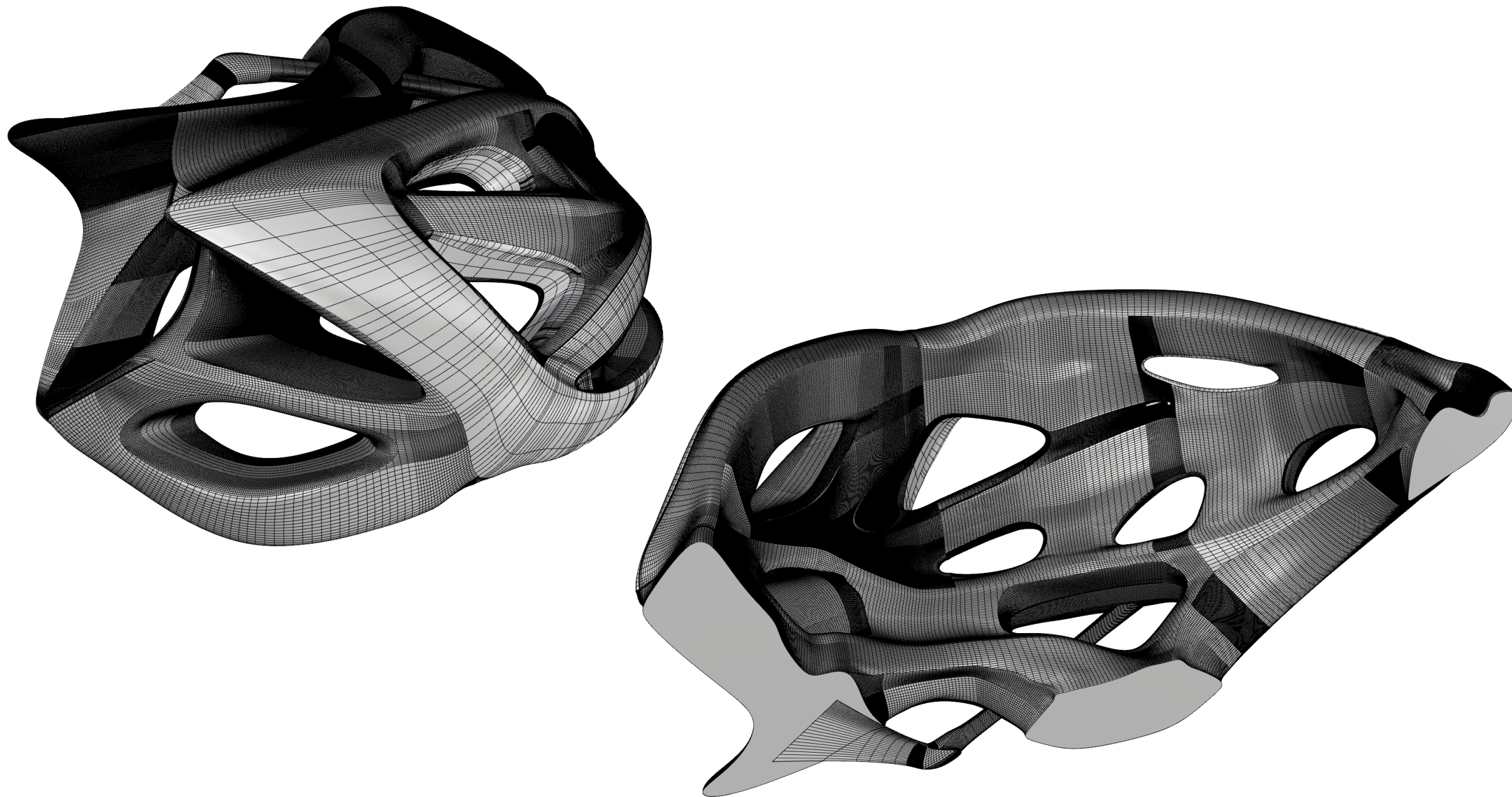
- To be able to join surfaces, $1e-3$ mm maximum deviation was required between the overlapping edges of two surfaces. This was checked with CrvDeviation. In case this was not achieved, MatchSrf with refine match was utilized.

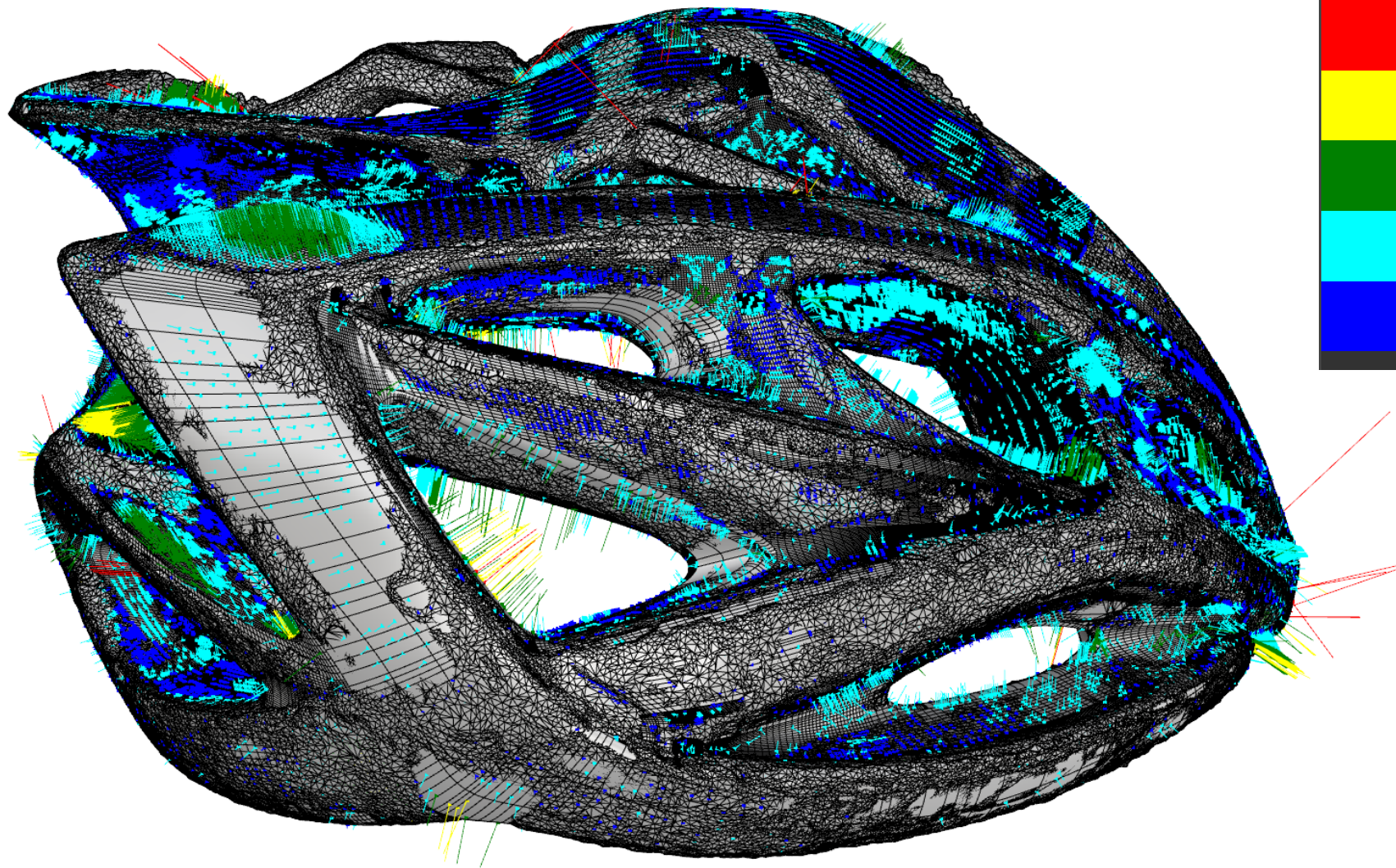
- For the representation of cooling holes on the NURBS surface:
 - For low-slope holes: the curves used to generate the NURBS surface were also representing the hole. After the generation of the surface, Project + Split were used to cut the surface.
 - For sharp transition holes: the outer surface was Split either with a PlanarSrf or with a Projected curve. Then, either the entire hole surface was modelled (NetworkSrf) or just a part of it (Loft or NetworkSrf), later connected to the outer surface with BlendSrf.



The PC insert was the same modelled previously.
The PC outer shells were extracted in the same way seen previously.





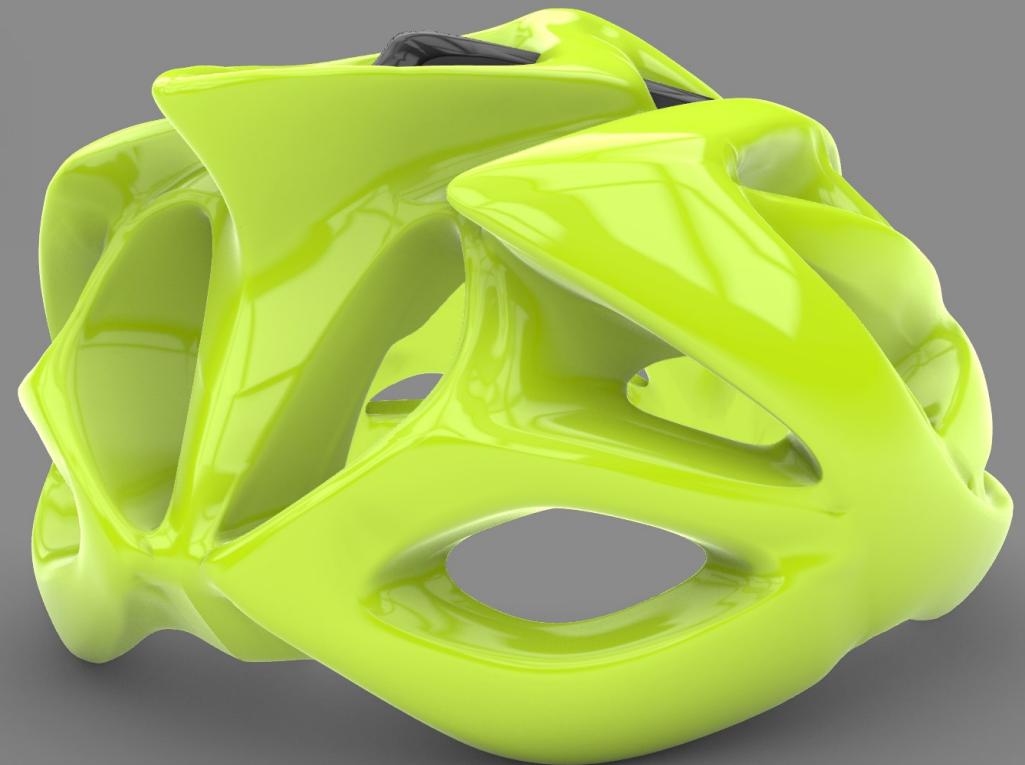


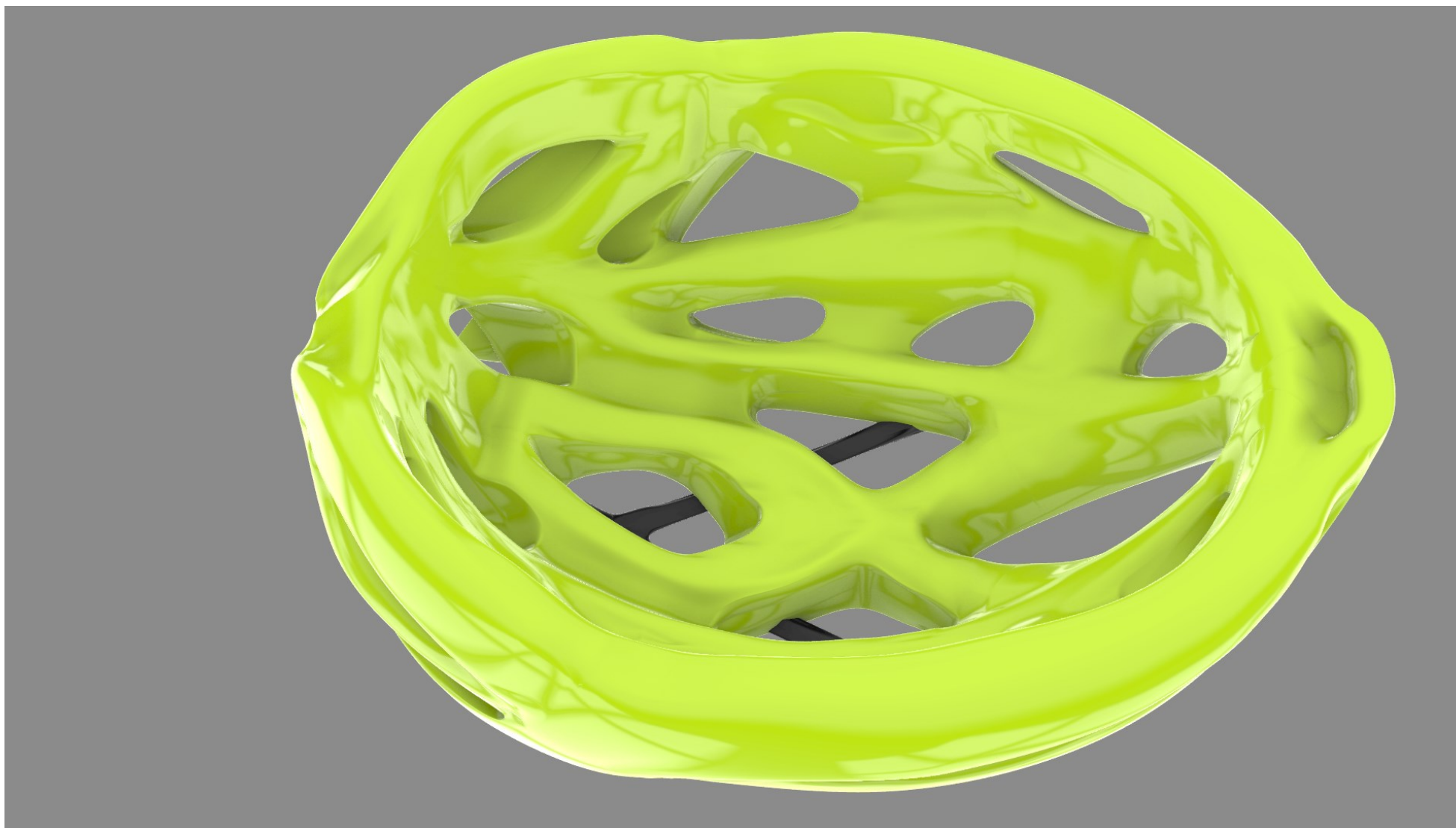
All measurements are in file units

	Ignore:	5
	Bad point:	1.5
	Good point:	0.1
	On Surface:	0

Statistics

Point count:	114314
Average distance:	0.17865
Median distance:	0.11493
Standard deviation:	0.23142
Maximum distance:	4.76840





Numerical results

04

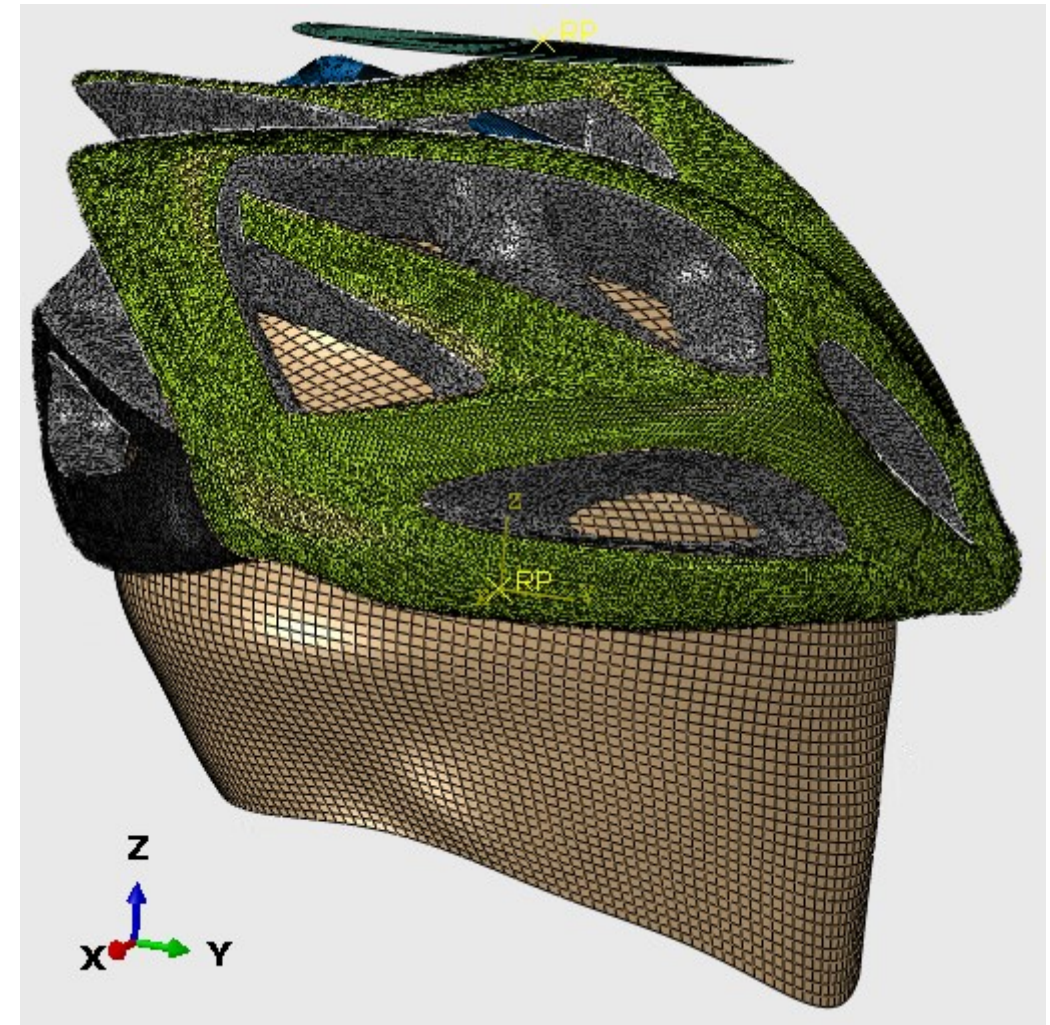
Abaqus (preprocessor) + OpenRadioss (solver)

- X-plane symmetry, given the central position of the anvil.
- Elastic perfectly plastic behaviour of PC (outer shells and insert).
- Linear elastic (with tensile cutoff stress) + plateau + densification for the EPS liner [2].
- Rigid headform and anvil.
- Node-to-surface contact pairs + tie constraints.
- Initial impact velocity: 5.42 m/s (EN 1078).
- Explicit dynamics with fixed time-step of $1.0\text{e-}7$ s and mass scaling.
- Simulated time: 12 ms.

Weight of the helmet: 215.0 g

Weight of the virtual helmet (Auto): 214.6 g (-0.2 %)

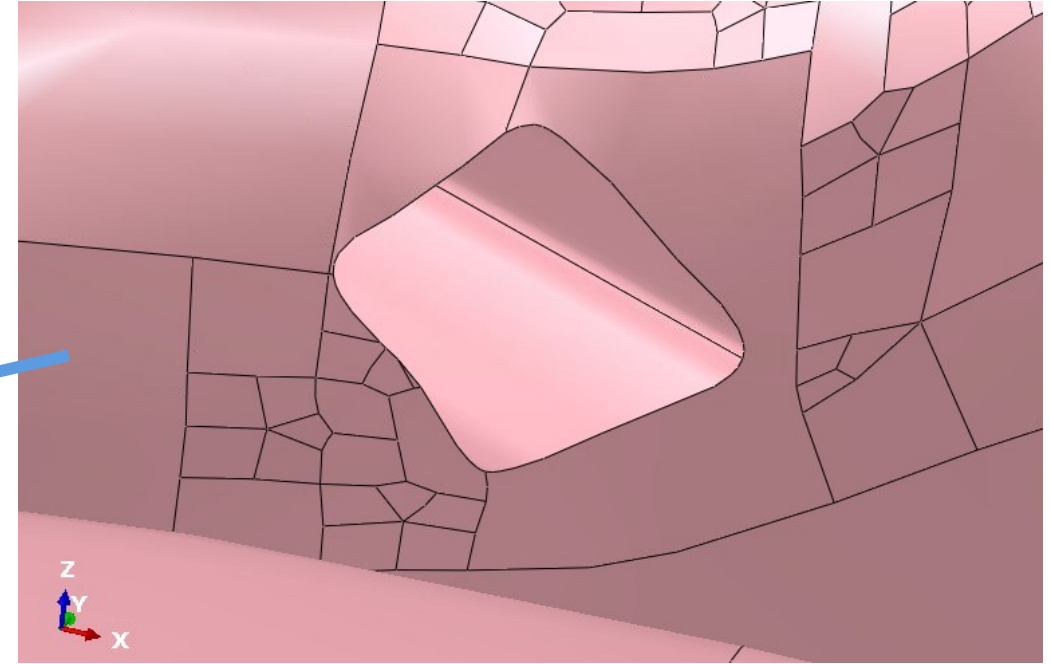
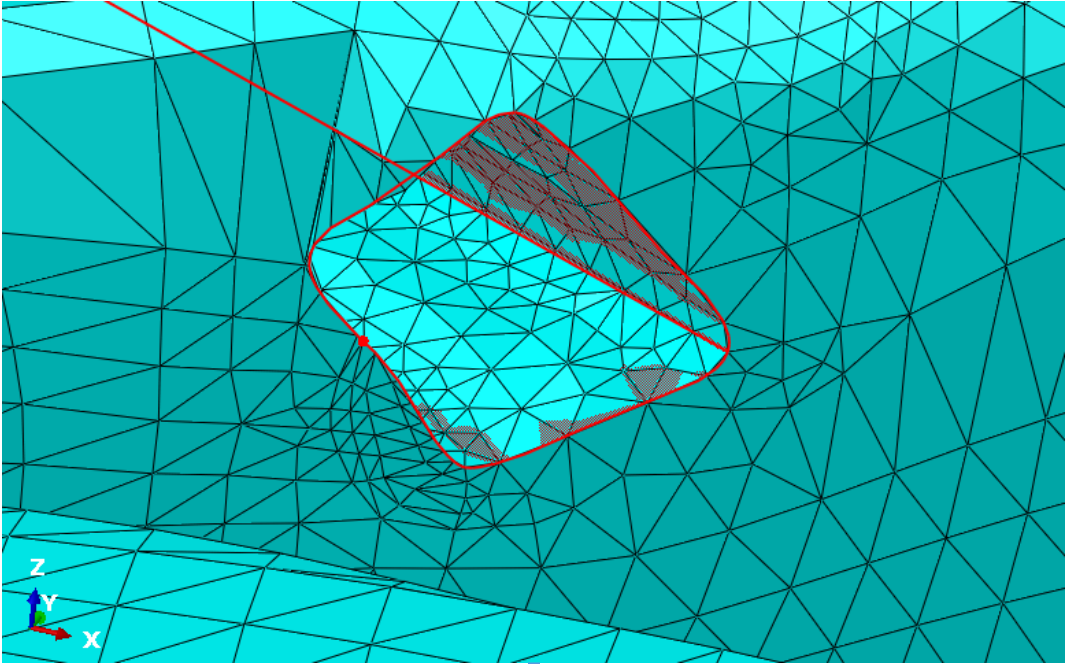
Weight of the virtual helmet (Manual): 224.6 g (+4.5 %)



Meshed assembly (manual surface modelling case) in Abaqus.

[2] Jiang J, Zhang Z, Zheng J, Shu K, Zhu W. Mechanical Properties Testing and Numerical Modeling and Simulations of a Nozzle Cover Made of Expanded Polystyrene. *Materials*. 2025; 18(16):3835. <https://doi.org/10.3390/ma18163835>

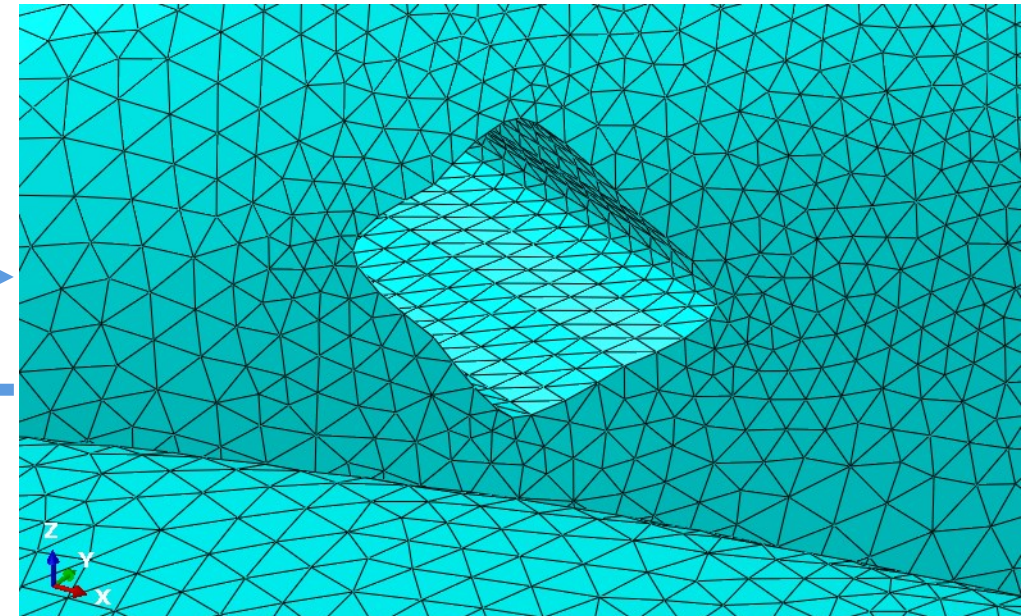
Mesh quality warnings (**Auto**): 173



After heavy use of virtual topology, ignoring 2966 entities

34 mesh quality warnings out of 1.06 M elements (0.0032%)

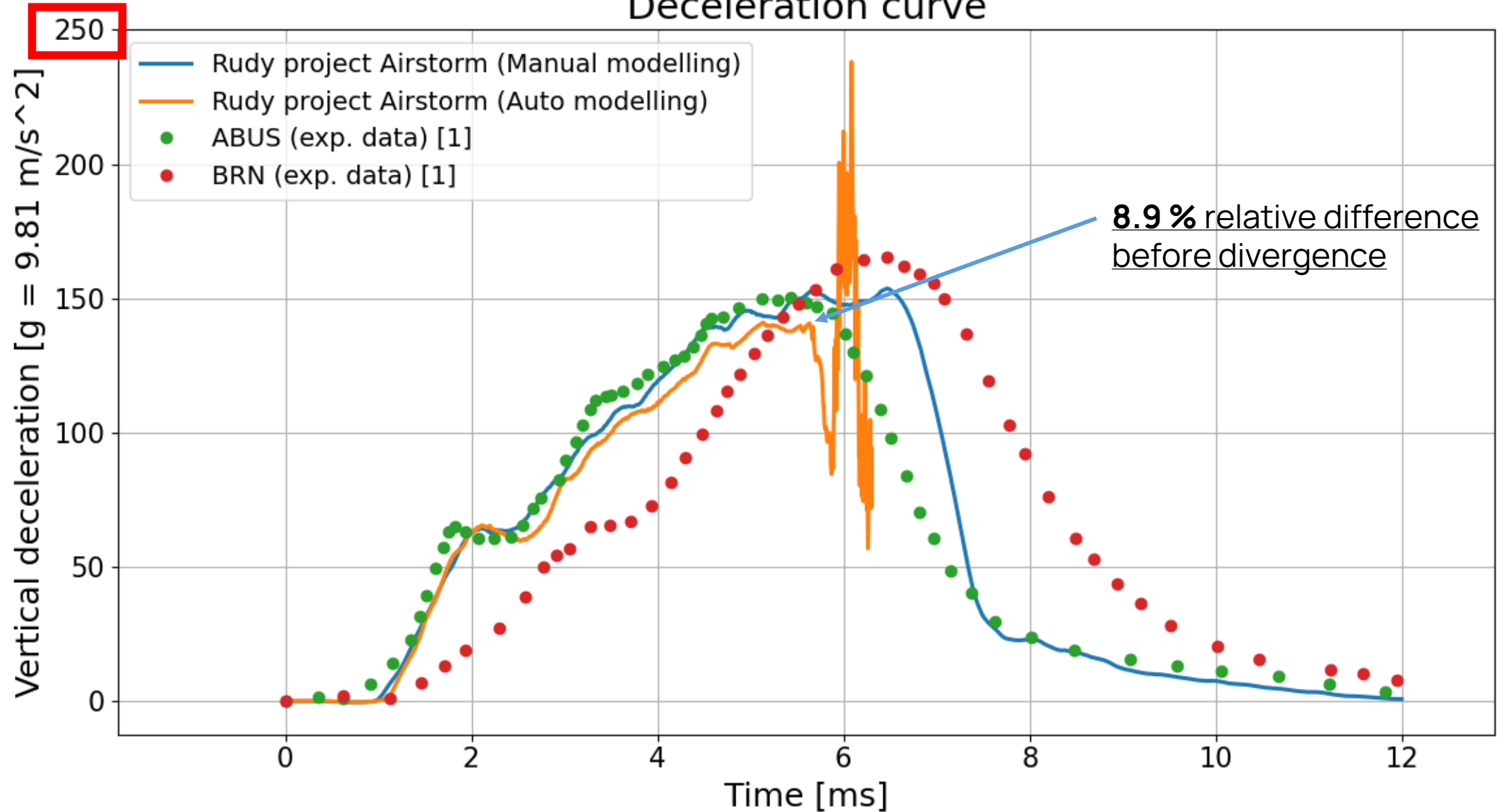
Mesh quality warnings (**Manual**): 31 out of 1.17M (0.0027%)



Comments:

- The simulation based on **auto** extracted NURBS surfaces and virtual topology diverged at around 0.4ms, caused by zero-energy modes close to the interface between the PC insert and the EPS liner.
- Element deletion helped reaching at most 5.5 ms but not without large errors in energy balance and mass scaling ($> 5\%$, rising steeply from the beginning).
- The simulation based on **manually** modelled NURBS surfaces, in the same settings as the previous one, wasn't prone to divergence. Energy balance error remained below 5%, same as for the mass scaling (max. +0.023%).

Deceleration curve



[1] Righetti E., (2024). Detailed FE simulations of helmet technologies and development of an experimental prototype for safety enhancement [Master's thesis, Politecnico di Milano].

Contact Energy comparison

